2026

Final Year Design Project

Policy, SOP, Evaluation & Workflow

FACULTY OF COMPUTING,
School of Electrical Engineering and Computer Science
National University of Sciences and Technology
Sector H-12
Islamabad

Revisions

Document Metadata

- Document Status: Approved
- Effective Date: Fall 2026

Version History

Version	Date	Description of Changes	Reviewed/Approved By
0.1	17-12-2025	Initial draft of FYDP evaluation SOP and stage-gate workflow.	Team
0.2	03-02-2026	Incorporated mathematical formulation for minimum and maximum faculty evaluation loads.	Team and HoDs
0.3	12-03-2026	Finalized G1–G4 milestone parameters. Submitted for Departmental Board of Studies review.	DBS
1.0	16-04-2026	Revisions based on DBS feedback regarding student group configurations.	DBS, Team
1.3	26-06-2026	Updated grading based on feedback from FBS	Team, FBS
1.4	27-07-2026	Revised governance framework based on committee feedback. Updated advisor roles, stage-gate governance, supervision policies, and evaluation procedures.	Team, DBS
1.5	01-10-2026	Updated work flow and timeline	FYDP Coordinators, HODs, DBS, Associate Dean FOC

Contents

1. Overview 1
2. Objectives 1
   a. Evaluation Structure 1
   b. Credit Hour Alignment 1
   c. Independent Academic Evaluation 1
   d. Industry Integration 1
   e. Advisor Governance Role 1
3. Stakeholders 2
4. Evaluation Structure and Grade Distribution 3
   a. Overall Evaluation Distribution 3
   b. Final Grade Computation 3
      i. Proposal Defense 4
      ii. Mid Defense: MVS + Project Documentation 4
      iii. Open House: Industrial Evaluation 4
      iv. Final Defense 5
5. Evaluation and Marks Allocation 5
   a. Marks Allocation 5
   b. FYDP Supervision and Co-Supervision Count 6
   c. Role of Evaluation Committee 6
   d. Role of Advisory and Co-advisor 6
6. Evaluation Outcomes 7
   a. Evaluation SoP 7
   b. Re-evaluation SOP for Failed Activities 8
7. Evaluation Rubric Structure 8
   a. Proposal Defense Rubrics 8
   b. Mid Defense Rubrics 8
   c. Industrial Evaluation Rubrics 9
   d. Final Defense Rubrics 9
   e. Complete division of marks and their weights 10
8. Final Year Design Project (FYDP) Framework 9

FYP Policy, SoP & Workflow

1. Overview

This document defines the evaluation structure for the two-semester Final Year Design Project (FYDP-I and FYDP-II). The evaluation framework is designed to simplify grading and strengthen independent academic evaluation. The revised SoP introduces four formal evaluation modules distributed across the two semesters and incorporates external industry assessment to strengthen real-world relevance.

2. Objectives

a. Evaluation Structure

The FYDP is evaluated through four major assessments:

1. Proposal Defense
2. Minimum Viable Solution (MVS) + Project Documentation Evaluation
3. Industrial Evaluation
4. Complete Project Defense

These assessments correspond to key technical maturity milestones in the project lifecycle.

b. Credit Hour Alignment

The evaluation model reflects the credit structure of the Final Year Design Project:

- FYDP-I: 2 Credit Hours
- FYDP-II: 4 Credit Hours

The weighted average of grades from all evaluation modules collectively determines the final FYDP grade.

c. Independent Academic Evaluation

Evaluation authority is shared between the advisor and the Evaluation Committee to ensure fairness, academic rigor, technical supervision, and cross-project standardization.

d. Industry Integration

External industry experts contribute to project evaluation to validate real-world relevance, deployment feasibility, and innovation impact.

e. Advisor Governance Role

In addition to assigning the evaluation marks allocated under this policy, the Advisor is responsible for ensuring project quality, validating deliverables, and monitoring technical progress.

3. Stakeholders

- **Students:** Final-year undergraduate students within the Faculty of Computing academic programs (BESE, BSCS, BSDS, and BSAI) who transition from their Junior year to Senior year upon becoming eligible to initiate the FYDP process.
- **Faculty Members (FM):** Faculty who can serve as advisors or evaluation committee members. Their role is to guide, supervise, or assess FYDP work through the TMS. Each FoC faculty member must choose the Knowledge Groups most relevant to their expertise by updating their profile on the TMS.
- **Advisor and Co-Advisor**: There is one compulsory Advisor and one optional Co-advisor. The Advisor should be a faculty member from SEECS. Any exception to this requirement shall be subject to approval by the relevant HoD or Associate Dean. The Co-advisor may be a faculty member from SEECS, another NUST department or school, or a qualified external academic or industry professional. The Advisor and Co-advisor (if any) supports the student team, monitors project progress, and helps ensure that the project is ready for evaluation.
- **Evaluation Committee:** A minimum two non-advisor faculty members that evaluate each project based on domain expertise.
- **External Industry Evaluators:** Subject matter experts from the industry who assess projects during the Open House.
- **Thesis Management System (TMS):** The centralized NUST digital portal for knowledge group management, project registration, submissions, and grade management.
- **FYDP Coordinator:** The administrative lead responsible for governance, timelines, and process oversight.
- **Program HoD / Associate Dean (AD):** The primary authority for conflict resolution.

4. Evaluation Structure and Grade Distribution

a. Overall Evaluation Distribution

The Final Year Design Project grade is determined through four evaluation modules distributed across the two semesters. The evaluation of Complete Project Defense will proceed once the Advisor has certified project readiness through the Health Check Report.

No.	Stage	Evaluation Timing	Weight	Purpose	Grading and CGPA
1	**Proposal Defense**	Week 3/4 of Semester 7	20%	Validate problem, baseline, feasibility	FYDP-I
2	**Mid Defense**: MVS + Project Documentation	Week 14/15 of Semester 7	30%	Validate MVS, SRS, architecture	13% to FYDP-I 17% to FYDP-II
3	**Open House**: Industrial Evaluation	Week 12 of Semester 8	10%	Validate industry relevance	FYDP-II
4	**Final Defense:** Complete Project Defense	Week 14/15 of Semester 8	40%	Validate full system and improvement	FYDP-II

b. Final Grade Computation

The final FYDP grade shall be calculated by normalizing the rubric scores at each evaluation and applying the prescribed evaluator weights. For the Proposal Defense, Mid Defense and Final Defense, the Advisor, co-advisor (if any), and Evaluation Committee shall contribute equally to the score. The Industrial Evaluators shall independently determine the Open House Industrial Evaluation score. The resulting contributions shall be aggregated to obtain the final grade out of 100.

No.	Evaluation	Advisor & co-advisor (if any)	Evaluation Committee	Industry Evaluators	Total
1	Proposal Defense	10%	10%	—	20%
2	**Mid Defense**: MVS + Documentation	15%	15%	—	30%
3	**Open House**: Industrial Evaluation	—	—	10%	10%
4	**Final Defense**	20%	20%	—	40%
	Total	**45%**	**45%**	**10%**	**100%**

For each evaluation stage, the score of the evaluation committee will be calculated in a 50:50 ratio. Additionally, when a Co-advisor is appointed, the score shall be calculated in a 60:40 ratio between the Advisor and Co-advisor.

Rubric marks represent raw assessment scores. These scores shall be normalized and converted into the prescribed weights using the calculation method defined above.

i. Proposal Defense

Timing: 2nd-3rd week of FYDP-I semester

This stage evaluates the intellectual foundation, feasibility, and methodological clarity of the proposed project. Evaluation focuses on:

- Problem definition and scope clarity
- Identification of baseline system or methodology
- Definition of measurable improvement
- Initial system architecture and workflow
- Technical feasibility and implementation plan
- Ethical considerations and AI compliance

Successful completion of this stage confirms readiness to proceed to system development.

ii. Mid Defense: MVS + Project Documentation

Timing: 14th – 15th week of FYDP-I semester

This evaluation validates the technical foundation and early implementation maturity of the project. Teams must demonstrate a **Minimum Viable Solution (MVS)** and submit structured technical documentation. Required documentation typically includes:

- Software Requirements Specification (SRS) and Literature Review
- System Design Specification (Architecture / SDS)
- Test Plan definitions
- Baseline implementation verification
- Initial module implementation
- Version-controlled source code repository (e.g., GitHub)

Evaluation focuses on:

- System architecture coherence
- Baseline implementation correctness
- MVS functionality
- Documentation quality and completeness
- Testing preparation and traceability

This stage ensures the project is technically viable and ready for full implementation during FYDP-II.

iii. Open House: Industrial Evaluation

Timing: 12th – 13rd week of FYDP-II Semester

Industry experts evaluate the practical impact and technology readiness of the project. Evaluation must occur during the Open House or an equivalent industrial review

session, where each evaluation must remain single blind. The panel of two/three Industry Evaluators shall determine the marks and evaluation outcome. The average of their individual marks shall constitute the final score, while the FYDP Coordinator shall formally record the marks and outcome in TMS.

Industry evaluation focuses on:

- Real-world applicability
- solution usability and scalability
- Innovation potential
- Deployment feasibility
- Industry relevance

The industrial evaluation contributes **10% of the total FYDP grade**.

iv. Final Defense

Timing: 14th – 15th week of FYDP-II Semester

This evaluation validates full system implementation, experimental validation, and measurable improvement over the baseline. Evaluation focuses on:

- Complete system functionality
- Integration maturity
- Demonstrated measurable improvement
- Evaluation rigor and benchmarking
- System stability and deployment readiness
- Documentation completeness and reproducibility
- Well-written code base and its understandability by all members of the FYDP Group.

Teams must present a working system and demonstrate the project through real-time technical demonstration rather than presentation-only slides.

5. Evaluation and Marks Allocation

a. Marks Allocation

The Advisor, Evaluation Committee (derived from KG Committee) and Industry Evaluators are responsible for timely, effective, and fair grading of all FYDPs within their purview. As such the evaluation is based on the following divisions:

- Advisor (and Co-advisor, if applicable): 45% total
- Evaluation Committee: 45%
- Industry Evaluators: 10%

Evaluation By	Members	Weight	Consolidation Criteria
Advisor (& Co-advisor)	1 Advisor, or 1 Advisor + 1 Co-Advisor	45%	Advisor only: full 45%. With Co-advisor: marks combined in **60:40 ratio (Advisor : Co-advisor)**.
Evaluation Committee (derived from KG Committee)	At least 2	45%	**Average** of the all evaluators' marks.
Industry Evaluators	3	10%	**Average** of individual marks

b. FYDP Supervision and Co-Supervision Count

A faculty member may supervise up to five projects as an advisor and five projects as a co-advisor. This arrangement provides all faculty members with an opportunity to participate in FYDP supervision. Any request to exceed these limits may be approved by the Associate Dean, Faculty of Computing, upon the recommendation of the relevant Head of Department. Diversity in advisor and co-advisor assignments is encouraged to promote broader collaboration among faculty members.

c. Role of Evaluation Committee

The Evaluation Committee serves as the independent academic evaluation body for FYDPs, alongside the assessment conducted by the Advisor and Co-advisor. The FYDP Coordinators shall constitute multiple Evaluation Committees from the available faculty members and assign projects to each committee through TMS.

The Proposal Defense, Mid Defense, and Final Defense shall each be conducted on a designated day. Every Evaluation Committee shall be allocated a specific venue where its assigned projects will be presented and evaluated. A Committee Head shall be appointed to ensure that at least two committee members are present during each evaluation and that the evaluation workload is distributed equitably among the members.

Each committee member shall be responsible for conducting academic evaluations; ensuring fairness and consistency across projects; verifying claims of measurable improvement; and assessing the technical rigor and quality of project documentation.

The Evaluation Committees and project assignments shall be finalized and uploaded to TMS by the second week of Semester 7. Students and faculty members shall be notified at least two weeks before the Proposal Defense. The same committees shall ordinarily evaluate their assigned projects at the Proposal, Mid, and Final Defense stages to ensure continuity and consistency.

d. Role of Advisory and Co-advisor

Each FYDP shall have one mandatory Advisor and may have one Co-advisor. The Advisor and Co-advisor shall guide the students, monitor technical progress, review deliverables, and assess the project's readiness for the Final Défense (FD). They shall also ensure compliance with the

University's regulations, SEECS documentation standards, plagiarism requirements, ethical principles, and the institutional policy on the use of artificial intelligence.

The Advisor shall be a regular faculty member of SEECS. Any exception shall require approval from the relevant HoD or Associate Dean. The Advisor shall have no personal, financial, or professional conflict of interest with the students; shall maintain confidentiality and respect intellectual property rights; and shall supervise projects within the approved workload limits. The Advisor shall receive supervision credit in accordance with the applicable NUST policy.

The Co-advisor may be a faculty member from any NUST school or department, or a suitably qualified academic or industry professional from a relevant field.

The Advisor and, where appointed, the Co-advisor shall:

- Assign evaluation marks within the prescribed timelines.
- Supervise the project's technical development and implementation.
- Verify the quality and maturity of the technical work.
- Validate the selected baseline and the proposed measurable improvements.
- Ensure that all required documentation meets SEECS standards.
- Monitor ethical compliance and the responsible use of artificial intelligence.
- Conduct plagiarism verification in accordance with the approved NUST policy.
- Verify the disclosure of AI-generated content and compliance with the applicable institutional policy.

6. Evaluation Outcomes

a. Evaluation SoP

The cumulative evaluation marks of the Proposal Défense (PD), Mid Défense (MD), and Final Défense (FD), shall determine and one of the following outcomes (based on pre-defined thresholds):

- **Approved:** The project meets the required academic and technical standards and may proceed to the next stage.
- **Approved with Observations:** Minor revisions are required. The Advisor shall ensure that all observations are addressed before the next evaluation.
- **Failed – Re-evaluation Required:** Major technical or methodological deficiencies have been identified. The team shall address these deficiencies and undergo one re-evaluation within the timeline approved by the relevant HoD. The marks obtained in the original evaluation shall be entered in TMS. In the case of MD, these marks shall contribute to the FYDP-I grade and may consequently affect the students' GPA. Re-evaluation is permitted only for PD and MD.
- **Rejected:** The project is fundamentally infeasible or academically invalid, has failed the permitted re-evaluation, or has failed the FD. The FYDP shall be terminated, and the students may register a new FYDP in accordance with the University regulations and with the approval of the relevant HoD.

For Industrial Evaluation, the outcome shall be determined by the panel of Industry Evaluators and formally recorded in TMS by the FYDP Coordinator.

b. Re-evaluation SOP for Failed Activities

An FYDP receiving a **Failed – Re-evaluation Required** outcome in PD or MD shall be permitted one re-evaluation. Unless otherwise approved by the relevant HoD, the re-evaluation shall be conducted within two weeks of the original evaluation. The FYDP team shall coordinate the re-evaluation with the Advisor, Co-advisor, where applicable, and the assigned Evaluation Committee.

If the FYDP fails the re-evaluation, the outcome shall be recorded as **Rejected**, and the project shall be terminated. The students may register a new FYDP in accordance with University regulations and applicable timelines, subject to the approval of the relevant HoD.

7. Evaluation Rubric Structure for Software Engineering (SE) Department

To ensure fair and consistent assessment across all evaluation stages, separate rubrics are provided for the four evaluation stages: Proposal Defense, Mid Defense, Open House and Final Defense. Each criterion is worth 5 marks.

Proposal Defense (PD) Rubric

Each criterion is marked out of 5 (score shown in the column header). PLO = Program Learning Outcome.

PLO	Criterion and evaluation question	0 – Not Demonstrated	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
2	**Problem Definition & Technical Context** *Is the problem clearly defined, appropriately scoped, and technically analyzed?*	Problem is not identified or defined.	Problem is unclear, overly broad, or lacks technical context.	Problem is defined but has gaps in scope, context, or analysis.	Problem is adequately defined and scoped with basic technical context.	Problem is clearly defined, appropriately scoped, and technically contextualized.	Problem is precisely formulated, well scoped, and comprehensively supported by technical/domain analysis.
4	**Baseline + Measurable Improvement Identification** *Are an appropriate baseline and measurable improvement targets defined for evaluating the proposed solution?*	No baseline or measurable improvement is defined.	Baseline or improvement target is unclear or inappropriate.	Baseline and basic metrics are identified but lack sufficient justification.	Appropriate baseline and basic measurable targets are defined.	Relevant baseline, metrics, and measurable improvement targets are clearly defined and justified.	Multiple relevant baselines and rigorous metrics with explicit, well-justified improvement targets are defined.
1	**Technical Solution Strategy** *Does the proposed solution apply relevant engineering and software engineering knowledge to address the problem?*	No viable technical solution is proposed.	Solution is largely inappropriate or technically weak.	Solution is partially feasible but has significant technical gaps.	Solution is feasible and addresses the problem with basic technical justification.	Solution is appropriate, feasible, and technically sound with good justification.	Solution demonstrates strong technical depth with well-justified and effective design choices.
3	**System Architecture / Methodology** *Is the proposed architecture and methodology clearly designed to address the problem and meet the requirements?*	No architecture or methodology is presented.	Architecture/methodology is unclear or inappropriate.	Basic architecture/methodology is presented but has significant gaps.	Architecture/methodology is adequately defined and addresses the main requirements.	Architecture/methodology is clear, coherent, and appropriate.	Architecture/methodology is comprehensive, technically rigorous, and thoroughly justified.
5	**Modern Tools & Technologies** *Are appropriate modern tools, technologies, frameworks, and techniques selected and justified?*	No relevant tools or technologies are identified.	Tools/technologies are inappropriate or unjustified.	Appropriate tools are identified but justification is weak.	Suitable tools and technologies are identified with basic justification.	Appropriate tools, technologies, frameworks, and techniques are clearly selected and justified.	Tools and technologies are systematically evaluated and selected based on technical requirements and project needs.
6	**Societal/Professional Considerations** *Does the project address a relevant societal problem and clearly identify its potential impact on society?*	No societal relevance or impact is identified.	Societal relevance is mentioned but the problem or potential impact is unclear.	A societal problem is identified, but its relevance or potential impact is weakly explained.	A relevant societal problem is identified and its potential impact is reasonably explained.	The societal problem is clearly established and the potential impact of the proposed solution is well explained.	The project addresses a significant societal problem, with its potential impact clearly articulated and supported by strong evidence or justification.
8	**Team Coordination & Collaboration** *Has the team demonstrated effective collaboration, role allocation, and coordination?*	No evidence of team coordination.	Roles and responsibilities are unclear.	Roles are defined but coordination or workload distribution has gaps.	Roles and responsibilities are adequately defined.	Roles, responsibilities, and coordination are clearly defined and appropriate.	Team structure, communication, coordination, and workload distribution demonstrate strong and balanced collaboration.
10	**Project Planning & Management** *Is the project realistically planned and managed using appropriate project management tools, development methodology, tasks, milestones, timeline, resources, and risk management?*	No meaningful project plan or management approach is presented.	Basic planning is presented, but tasks, timeline, responsibilities, or management approach are unclear. No appropriate project management tool or methodology is identified.	A basic plan and development methodology are identified, but important elements such as milestones, resources, risks, or project management tools are inadequately addressed.	A reasonable project plan is presented with defined tasks, milestones, timeline, and development methodology. Appropriate project management tools are identified.	The project is realistically planned using appropriate project management tools and methodology, with clear tasks, milestones, responsibilities, resources, timeline, and risk management.	The project demonstrates a comprehensive and well-justified management approach, with effective integration of project management tools, development methodology, task tracking, milestones, dependencies, resources, timeline, and risk management.

PLO	Criterion and evaluation question	0 – Not Demonstrated	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
9	**Technical Presentation & Communication** *Does the team clearly present the proposed work and effectively respond to technical questions?*	Not presented or cannot adequately communicate the proposed work.	Presentation is unclear and responses show limited understanding.	Main ideas are communicated but with significant gaps in clarity or technical responses.	Proposed work is adequately communicated and basic questions are addressed.	Presentation is clear and well structured, with appropriate responses to technical questions.	Communication is clear, precise, technically strong, and demonstrates thorough understanding of the proposed work.

Open House Rubric

Each criterion is marked out of 5 (score shown in the column header). PLO = Program Learning Outcome.

PLO	Criterion and evaluation question	0 – Not Demonstrated	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
2	**Industry Relevance & Practical Utility** *Does the solution address a meaningful real-world or industry problem and provide practical value to its intended users or stakeholders?*	No meaningful real-world or industry problem is demonstrated.	Problem relevance and practical value are very limited.	Problem is moderately relevant, but practical value is insufficiently demonstrated.	Solution addresses a relevant practical problem and provides clear value.	Solution addresses a significant practical or industry need and demonstrates clear value.	Solution addresses a significant real-world or industry need with strong potential for meaningful practical impact.
3	**Deployment Feasibility & Solution Potential** *Is the solution technically and operationally feasible for commercial deployment, adoption, or scalability?*	Solution is not demonstrably feasible for practical use.	Feasibility is very limited; substantial development is required.	Basic feasibility is demonstrated, but significant barriers remain.	Solution is feasible for its intended context with reasonable further work.	Solution demonstrates strong feasibility for deployment, adoption, or further development.	Solution demonstrates a credible pathway to deployment, adoption, or scaling with strong potential for sustained practical impact.

Mid Defense (MD) Rubric

Each criterion is marked out of 5 (score shown in the column header). PLO = Program Learning Outcome.

PLO	Criterion and evaluation question	0 – Not Demonstrated	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
2	**Requirements Engineering (FR/NFR)** *Are the functional and non-functional requirements complete, clear, verifiable, and traceable to the project objectives?*	Requirements are absent or cannot be meaningfully evaluated.	Requirements are incomplete, unclear, or substantially inconsistent.	Basic functional and non-functional requirements identified, with significant gaps.	Functional and non-functional requirements clearly defined with reasonable coverage.	Requirements are structured, verifiable, and traceable to objectives or system components.	Requirements are comprehensive, prioritized, verifiable, and fully traceable to objectives, design, and implementation.
3	**System Design and Architectural Pattern** *Does the proposed system architecture provide a coherent and appropriate structure for implementing the identified requirements, with appropriate selection and justification of relevant architectural/design patterns?*	No meaningful system architecture or design is presented.	Design is incomplete, unclear, or technically unsuitable.	Basic architecture presented, but components, interactions, or decisions lack detail or justification.	Components, interactions, data flow, and design decisions clearly align with requirements.	Architecture is modular and justified, with appropriate patterns supporting separation of concerns and integration.	Architecture is robust and well justified, addressing relevant scalability, maintainability, extensibility, performance, security, and integration constraints.
4	**Test Plan & Evaluation Strategy** *Does the project have a systematic plan to test and evaluate the implemented solution using appropriate test cases, metrics, and criteria?*	No meaningful testing or evaluation plan is presented.	Testing mentioned without clear objectives, scope, or test cases.	Basic test cases or evaluation objectives identified, with significant omissions.	Structured test plan defines objectives, scope, test cases, and evaluation measures.	Actionable test plan includes appropriate tools, measurable criteria, and requirement traceability.	Comprehensive evaluation strategy integrates coverage, rigorous metrics, traceability, risks, and a realistic execution plan.
5	**Code Quality, Structure & Development Practices** *Does the implemented solution demonstrate well-structured, modular, readable, and maintainable code, supported by appropriate development practices and tools?*	No usable codebase or development evidence is available.	Code is poorly organized and development practices are largely absent.	Basic code organization and version control are present but inconsistently applied.	Code is structured and version-controlled, with appropriate development tools and practices.	Code demonstrates modularity, maintainability, documentation, and effective tool use.	Professional engineering workflows consistently support maintainable code, disciplined version control, and appropriate tools and frameworks.
9	**Technical Documentation & Traceability** *Is the technical documentation complete, well structured, and consistent and traceable across requirements, design, implementation, and testing?*	Required technical documentation is absent or unusable.	Documentation is incomplete, poorly structured, or inconsistent.	Basic documentation available, with significant omissions.	Documentation is structured and reflects the current system.	Documentation is complete, technically sound, and consistent with requirements, design, and implementation.	Documentation is rigorous, traceable across project artifacts, and supports reproducibility and continued development.
10	**Technical Feasibility & Project Execution** *Is the proposed solution technically feasible and cost-effective, with appropriate resource utilization, task allocation, and planning for project completion?*	No feasibility or execution plan evident.	Infeasible solution with poor planning.	Basic planning with limited justification of cost-effectiveness.	Feasible solution with reasonable resource use and completion planning.	Justified, cost-effective solution with effective resource and task management.	Well-justified, cost-effective solution with optimized resources, clear responsibilities, and realistic completion planning.

Final Defense (FD) Rubric

Each criterion is marked out of 5 (score shown in the column header). PLO = Program Learning Outcome.

PLO	Criterion and evaluation question	0 – Not Demonstrated	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
3	**System Implementation Completeness** *To what extent has the proposed solution been implemented and demonstrated against its intended functionality?*	No meaningful implementation or demonstration.	Few components implemented; major functionality missing.	Core functionality partly implemented; major gaps remain.	Major functionality implemented and demonstrated.	Components integrated and functioning coherently.	Complete, robust solution demonstrated against requirements.
1	**Technical Depth of Implementation** *Can the student(s) clearly explain and justify the technical concepts, components, technologies, and implementation decisions underlying the solution?*	Cannot explain the solution or its concepts.	Shows superficial understanding.	Explains basic components but has significant gaps.	Clearly explains key concepts, technologies, and decisions.	Explains components in detail and justifies decisions.	Demonstrates comprehensive understanding and sound technical justification.
4	**Experimental Evaluation & Improvement over Baseline** *Does the project provide systematic experimental evidence of performance and improvement over an appropriate baseline?*	No meaningful evaluation or baseline comparison.	Weak evaluation with unreliable evidence.	Basic evaluation; limited metrics, analysis, or comparisons.	Appropriate experiments, metrics, and baseline demonstrate improvement.	Strong validation using relevant data, metrics, and analysis.	Systematic benchmarking, detailed analysis, and statistical validation where applicable.
5	**Code Quality & Reproducibility** *Does the implementation demonstrate appropriate use of modern development tools and practices, with code and artifacts that are structured, documented, maintainable, and reproducible?*	Source code or essential artifacts unavailable.	Code is incomplete, poorly organized, or undocumented.	Code is available but has limited documentation, version control, or reproducibility.	Code is documented, version-controlled, and reusable.	Code is structured and maintainable, with documented dependencies and configurations.	Code is fully reproducible, with documentation, scripts, version history, and reproduction instructions.
7	**Ethical & Professional Integrity** *Do student(s) demonstrate ethical and professional integrity through ownership, attribution, and transparent disclosure of contributions, resources, tools, and limitations?*	No ownership; substantial work misrepresented	Minimal ownership or disclosure of contributions and resources.	Partial ownership with significant disclosure gaps.	Adequate ownership; contributions/resources appropriately disclosed.	Work ownership and external resources adequately disclosed.	Full ownership, transparent attribution, and responsible disclosure of AI tools and limitations.
8	**Individual & Team Contribution** *Does each team member demonstrate meaningful individual contribution, effective collaboration, and sufficient understanding of the overall project?*	Team members cannot demonstrate meaningful individual contribution or understanding of the project.	Contributions are uneven; understanding is limited.	Partial contributions and understanding; coordination is weak.	Each member contributes meaningfully and understands assigned work.	Strong individual contributions, coordination, and shared understanding.	Substantial individual contributions, excellent collaboration, and comprehensive team understanding.
9	**Communication & Defense** *Does the team communicate the technical work effectively through the presentation, demonstration, documentation, and responses to questions during the defense?*	The team cannot effectively present, demonstrate, or explain the project.	Presentation is unclear; responses show limited understanding.	Basic presentation and demonstration, with gaps in clarity or completeness.	Team clearly communicated the problem, solution, results, and key decisions.	Team delivered a coherent demonstration and confidently answered technical questions.	Team communicated complex issues professionally and answered difficult questions convincingly.
11	**Lifelong Learning & Technological Awareness** *Does the team demonstrate awareness of relevant technological developments, limitations of the current solution, and opportunities for continued learning and improvement?*	No awareness of relevant technological developments or opportunities for further learning is demonstrated.	Limited awareness of relevant technologies, developments, or limitations of the current solution.	Some relevant technologies or future improvements are identified, but understanding is superficial.	Relevant developments and learning opportunities identified	Emerging technologies, limitations, and meaningful improvements evaluated	Technological trends critically assessed and well-grounded directions for innovation and continued learning are proposed.

PLO	Criterion and evaluation question	0 – Not Demonstrated	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
6	**Societal, Safety, Legal & Cultural Considerations** *Does the solution appropriately address relevant societal, health, safety, legal, and cultural considerations and the associated responsibilities of an engineer?*	No relevant societal, safety, legal, or cultural considerations are identified.	Relevant considerations are largely overlooked or misunderstood.	Some relevant issues are identified, but their implications are inadequately addressed.	Relevant considerations are appropriately identified and addressed.	Thoughtful consideration of relevant implications and responsibilities, with appropriate mitigation where required.	Relevant implications are systematically evaluated and incorporated into the solution and its intended use.

8. Final Year Design Project (FYDP) Framework

Phase	Activity	Deadline	Responsibility
Pre-FYDP	Student registration in TMS	Week 14, Spring (Junior Year)	Coordinator + TMS
	Faculty profile update in TMS (Selection of relevant Knowledge Groups)	Week 1, Summer Break	Faculty Members (FM) + TMS
	Advisor-Student matching & registration	Week 2, Summer Break	Advisor + Students
	Resolution of unassigned groups/FMs	Week 3, Summer Break	FYDP Coordinator + TMS
	Final group setup & threshold validation	Week 5, Summer Break	FYDP Coordinator
	Evaluation Committee formation	Week 6, Summer Break	FYDP Coordinator
FYDP-1	Requirement Engineering phase	Week 1, Semester 7	Students
	Proposal Defense (PD) documentation submission and review	Week 1, Semester 7	Students + Advisor + co-advisor (if any)
	PD Presentation Activity	Week 2- 3 Semester 7	All Hands
	PD Grade Entry	Within three working days after the Proposal Defense	Evaluation Committee + Advisor + co-advisor (if any) + TMS
	Re-Evaluation Activity of PD	Week 4-5, Semester 7	Advisor + co-advisor (if any)+ Students
	Mid-Defense (MD) MVS & document submission and review	Week 13, Semester 7	Students + Advisor + co-advisor (if any)
	Mid-Defense Presentation Activity	Week 14-15, Semester 7	All Hands
	Mid-Defense Grade Entry	Within three working days after the Mid-Defense	Evaluation Committee + Advisor + co-advisor (if any)+ TMS
FYDP-2	Re-Evaluation Activity of Mid Defense	Week 1-2, Semester 8	Advisor + co-advisor (if any)+ Students
	Advisor Health Check	Week 11, Semester 8	Advisor
	Open House	Week 12 or 13, Semester 8	Industry + Students
	Final Defense (FD) Document Submission, Project Deployment	Week 13, Semester 8	Students

	FD Presentation Activity	Week 14-15, Sem 8	All Hands
	Final Grade Entry & Compilation	Week 17–18, Sem 8	Evaluation Committee + Advisor + co-advisor (if any)+ Coordinators

**Scheduling Note: The exact dates of all evaluation activities shall be notified by the FYDP Coordinator according to the approved academic calendar. Marks shall be entered in TMS within three working days after the evaluation.*